MERCK MANUAL
Professional Version

Hypertension

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Hypertension is sustained elevation of resting systolic blood pressure (≥ 130 mm Hg), diastolic blood pressure (≥ 80 mm Hg), or both. Hypertension with no known cause (primary hypertension) is most common. Hypertension with an identified cause (secondary hypertension) is usually due to primary aldosteronism. Sleep apnea, chronic kidney disease, or renal artery stenosis are other causes of secondary hypertension. Usually, no symptoms develop unless hypertension is severe or long-standing, which results in target organ damage to the brain, heart, or kidneys. Diagnosis is by sphygmomanometry. Tests may be done to determine cause, assess organ damage, and identify other cardiovascular risk factors. Treatment involves lifestyle changes and medications, including diuretics, angiotensin-converting enzyme (ACE) inhibitors, angiotensin II receptor blockers, and calcium channel blockers.

Hypertension is defined as a systolic blood pressure (BP) ≥ 130 mm Hg or a diastolic blood pressure ≥ 80 mm Hg or taking medication for hypertension. This definition is based on the relationship between blood pressure and cardiovascular events in large populations. Nearly half of adults in the United States have hypertension. Many of these people are not aware that they have hypertension. Approximately 80% of adults with hypertension have been advised to begin treatment with medication and lifestyle modification, but only approximately 50% with hypertension actually receive treatment (1).

Even with medication and lifestyle modification, BP is at goal (under control) in only 26% of patients, and, among treated adults whose BP is not at goal, almost 60% have a BP ≥ 140/90 mm Hg (1).

High blood pressure is more common in non-Hispanic Black adults (58%) than in non-Hispanic White adults (49%), non-Hispanic Asian adults (45%), or Hispanic adults (39%) (1). Among those for whom blood pressure medication and lifestyle modification have been recommended, control of blood pressure is higher among non-Hispanic White adults (31%) than in non-Hispanic Black adults (20%), non-Hispanic Asian adults (24%), or Hispanic adults (23%) (1).

Blood pressure increases with age. About two-thirds of people > 65 years have hypertension, and people with a normal BP at age 55 have a 90% lifetime risk of developing hypertension (2). Although hypertension is common with increased age, treatment remains important because higher BP increases morbidity and mortality risk.

Special considerations are indicated for hypertension during pregnancy because complications are different. Hypertension that develops during pregnancy may resolve after pregnancy (see Hypertension in Pregnancy and Preeclampsia and Eclampsia).

Categories of BP in adults, defined by the American College of Cardiology/American Heart Association (ACC/AHA), include normal, elevated BP, stage 1 (mild) or stage 2 hypertension (3) (see table Classification of Blood Pressure in Adults). Normal values for blood pressure in infants, children, and adolescents are much lower than those in adults (4).

TABLE

Classification of Blood Pressure in Adults*

Classification	Blood Pressure
Normal blood pressure	< 120/80 mm Hg
Elevated blood pressure	120–129/< 80 mm Hg
Stage 1 hypertension	130–139 mm Hg (systolic) OR 80–89 mm Hg (diastolic)
Stage 2 hypertension	≥ 140 mm Hg (systolic) OR ≥ 90 mm Hg (diastolic)

* Patients with systolic and diastolic blood pressure in different categories should be designated to the higher blood pressure category.

Data from Writing Committee Members*, Jones DW, Ferdinand KC, et al. 2025 AHA/ACC/AANP/AAPA/ABC/ACCP/ACPM/AGS/AMA/ASPC/NMA/PCNA/SGIM Guideline for the Prevention, Detection, Evaluation and Management of High Blood Pressure in Adults: A Report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Hypertension.* 2025;82(10):e212-e316. doi:10.1161/HYP.0000000000000249

General references

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Etiology of Hypertension

Hypertension Myths

PODCAST

Hypertension may be:

- Primary (no specific cause—85% of cases)
- Secondary (an identified cause)

Primary hypertension

Hemodynamics and physiologic components (eg, plasma volume, activity of the renin-angiotensin system) vary, indicating that primary hypertension is unlikely to have a single cause. Even if one factor is recognized, multiple factors are probably involved in sustaining elevated blood pressure (the mosaic theory). Heredity is a predisposing factor, but the exact mechanism by which genetics plays a role is unclear. At younger ages, environmental factors (eg, dietary sodium, stress) seem to affect only people who are genetically susceptible; however, in patients > 65 years, high sodium intake becomes more likely to precipitate hypertension.

Secondary hypertension

Common causes include (1):

- Obstructive sleep apnea
- Renal parenchymal disease: Chronic glomerulonephritis or pyelonephritis, polycystic renal disease, lupus nephritis, obstructive uropathy
- Renovascular disease
- Primary aldosteronism (2)
- Other endocrine disorders (less common): Pheochromocytoma, Cushing syndrome, congenital adrenal hyperplasia, hyperthyroidism, hypothyroidism (myxedema), primary hyperparathyroidism, acromegaly, and mineralocorticoid excess syndromes other than primary aldosteronism
- Prescription or over-the-counter medications: Sympathomimetics (decongestants), nonsteroidal anti-inflammatory drugs (NSAIDs), antipsychotics, tyrosine kinase inhibitors, angiogenesis inhibitors, stimulants, corticosteroids, cocaine
- Vascular anomalies: Coarctation of the aorta

Licorice may contribute to worsening of blood pressure control. Diabetes is not considered a cause of secondary hypertension, although patients with diabetes commonly also are hypertensive.

Etiology references

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Pathophysiology of Hypertension

Because blood pressure equals cardiac output (CO) multiplied by total peripheral vascular resistance (TPR), pathogenic mechanisms involve:

- Increased CO
- Increased TPR
- Both

In most patients with hypertension, CO is normal or slightly increased, and TPR is increased. This pattern is typical of primary hypertension and hypertension due to primary aldosteronism, pheochromocytoma, renovascular disease, and renal parenchymal disease.

In other patients, CO is increased (possibly because of venoconstriction in large veins), and TPR is inappropriately normal for the level of CO. Later in the disorder, TPR increases and CO returns to normal, probably because of autoregulation.

Some disorders that increase CO (eg, thyrotoxicosis, arteriovenous fistula, aortic regurgitation), particularly when stroke volume is increased, cause isolated systolic hypertension. Some older patients have isolated systolic hypertension with normal or low CO, probably due to inelasticity of the aorta and its major branches. Patients with high, fixed diastolic pressures often have decreased CO.

Plasma volume tends to decrease as BP increases; rarely, plasma volume remains normal or increases. Plasma volume tends to be high in hypertension due to primary aldosteronism or renal parenchymal disease and may be quite low in hypertension due to pheochromocytoma.

Abnormal sodium transport

In many cases of hypertension, sodium transport across the vascular cell membrane is abnormal, because the sodium-potassium pump (Na^+, K^+-ATPase) is defective or inhibited, or because permeability to sodium ions is increased. The result is increased intracellular sodium and calcium, which makes the cell more sensitive to sympathetic stimulation. Because Na^+, K^+-ATPase may pump norepinephrine back into sympathetic neurons (thus inactivating this neurotransmitter), inhibition of this mechanism could also enhance the effect of norepinephrine, increasing BP. Defects in sodium transport may occur in children who are normotensive but have a parent with hypertension.

Sympathetic nervous system

Sympathetic stimulation increases blood pressure, usually more in patients with hypertension than in patients who are normotensive. Whether this hyperresponsiveness resides in the sympathetic nervous system or in the myocardium and vascular smooth muscle is unknown.

A high resting pulse rate, which may result from increased sympathetic nervous activity, is a well-known predictor of hypertension.

In some patients with hypertension, circulating plasma catecholamine levels during rest are higher than normal.

Renin-angiotensin-aldosterone system

The renin-angiotensin-aldosterone system helps regulate blood volume and therefore blood pressure. Renin, an enzyme formed in the juxtaglomerular apparatus, catalyzes conversion of angiotensinogen to angiotensin I. This inactive product is cleaved by angiotensin-converting enzyme (ACE), mainly in the lungs but also in the kidneys and brain, to angiotensin II, a potent vasoconstrictor that also stimulates autonomic centers in the brain to increase sympathetic discharge and stimulates release of aldosterone and vasopressin. Aldosterone causes sodium retention and vasopressin causes water retention, elevating BP. Aldosterone also enhances potassium excretion; low plasma potassium (< 3.5 mEq/L [< 3.5 mmol/L]) increases vasoconstriction through closure of potassium channels.

Renin secretion is controlled by at least 4 mechanisms, which are not mutually exclusive:

- A renal vascular receptor responds to changes in tension in the afferent arteriolar wall
- A macula densa receptor detects changes in the delivery rate or concentration of sodium chloride in the distal tubule
- Circulating angiotensin II has a negative feedback effect on renin secretion

- Sympathetic nervous system stimulates renin secretion mediated by beta-receptors (via the renal nerve)

Elevated angiotensin II levels are generally acknowledged to be responsible for renovascular hypertension, at least in the early phase, but the role of the renin-angiotensin-aldosterone system in primary hypertension is not established. However, in patients with African ancestry and older patients with hypertension, renin levels tend to be low (1). Older patients also tend to have low angiotensin II levels, which may be related to a more volume-dependent salt-sensitive form of hypertension.

Hypertension due to chronic renal parenchymal disease (renoprival hypertension) results from the combination of a renin-dependent mechanism and a volume-dependent mechanism. In most cases, increased plasma renin activity is not evident. Hypertension is typically moderate and sensitive to sodium and water balance.

Vasodilator deficiency

Deficiency of a vasodilator (eg, bradykinin, nitric oxide), rather than excess of a vasoconstrictor (eg, angiotensin, norepinephrine), may cause hypertension. Reduction in nitric oxide, due to stiff arteries, occurs with aging, and this reduction contributes to salt sensitivity (ie, lesser amounts of salt ingestion will raise BP) (2). With reduced nitric oxide, a large sodium load (eg, a salty meal) may increase systolic BP by > 10 to 20 mm Hg.

If the kidneys do not produce adequate amounts of vasodilators (because of renal parenchymal disease), blood pressure can increase.

Vasodilators and vasoconstrictors (mainly endothelin) are also produced in endothelial cells. Therefore, endothelial dysfunction greatly affects blood pressure.

Pathology and complications

No pathologic changes occur early in hypertension. Severe or prolonged hypertension damages target organs (primarily the cardiovascular system, brain, and kidneys), increasing risk of:

- Coronary artery disease (CAD) and myocardial infarction (MI)
- Heart failure
- Stroke (particularly hemorrhagic)
- Renal failure
- Death

Elevated BP levels lead to the development of generalized arteriolosclerosis and acceleration of atherogenesis. Arteriolosclerosis is characterized by medial hypertrophy, hyperplasia, and hyalinization of small arteries (arterioles), notably in the eyes and the kidneys. In the kidneys, the changes narrow the arteriolar lumen, increasing TPR; thus, hypertension leads to more hypertension. Furthermore, once arteries are narrowed, any slight additional shortening of already hypertrophied smooth muscle reduces the lumen to a greater extent than in normal-diameter arteries. These effects may explain why the longer hypertension has existed, the less likely specific treatment (eg, renovascular surgery) for secondary causes is to restore blood pressure to normal. Thus, earlier diagnosis and treatment of even modestly elevated BP (ie, systolic BP of 120 to 129 mm Hg, even if the diastolic BP is in normal range) may provide benefit, especially in younger patients with a family history of hypertension.

Because afterload is increased in patients with elevated BP, the left ventricle gradually hypertrophies, causing diastolic dysfunction. The ventricle eventually dilates, causing dilated cardiomyopathy and heart failure, often worsened by arteriosclerotic coronary artery disease. Thoracic aortic dissection is typically a consequence of hypertension; almost all patients with abdominal aortic aneurysms have hypertension.

Pathophysiology references

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Symptoms and Signs of Hypertension

Hypertension is usually asymptomatic until complications develop in target organs. Even severe hypertension (typically defined as systolic blood pressure ≥ 180 mm Hg and/or diastolic blood pressure ≥ 120 mm Hg) can be asymptomatic (hypertensive urgency). When severe hypertension causes cardiovascular, neurologic, renal, and retinal symptoms (eg, symptomatic coronary atherosclerosis, heart failure, hypertensive encephalopathy, renal failure), it is referred to as a hypertensive emergency.

Retinal changes may include arteriolar narrowing, hemorrhages, exudates and, in patients with encephalopathy, papilledema (hypertensive retinopathy). Changes are classified (according to the Keith-Wagener-Barker classification) into 4 groups with increasingly worse prognosis (1):

- Grade 1: Constriction of arterioles only
- Grade 2: Constriction and sclerosis of arterioles
- Grade 3: Hemorrhages and exudates in addition to vascular changes
- Grade 4: Papilledema

Symptoms and signs reference

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Diagnosis of Hypertension

- Multiple measurements of BP
- Testing to diagnose causes and complications

Hypertension is diagnosed by sphygmomanometry. History, physical examination, and other tests help identify etiology and determine whether target organs are damaged.

Multiple measurements of BP are needed to confirm hypertension because of the inherent variability of blood pressure. Blood pressure typically fluctuates with time of day; in classic diurnal variation, BP is higher by day (especially in the morning) and lower by night. Home blood pressure measurement, using a cuffed device, can help confirm a diagnosis of hypertension (1).

Blood pressure measurement

The blood pressure used for formal diagnosis should be an average of 2 or 3 measurements taken in the office at different times under these conditions:

- Patient seated in a chair (not examination table) for > 5 minutes, feet on floor, back supported
- Upper limb supported at heart level with no clothing covering the area of cuff placement
- No exercise, caffeine, or tobacco use for at least 30 minutes

At the first visit, measure BP in both arms; subsequent measurements should use the arm that gave the higher reading.

A **properly sized BP cuff** is applied to the upper arm. An appropriately sized cuff covers two-thirds of the biceps; the bladder is long enough to encircle > 80% of the arm, and bladder width equals at least 40% of the arm's circumference. Thus, patients with obesity usually require large cuffs.

Blood pressure may be measured by either auscultatory or oscillometric devices. Fully automated validated oscillometric devices may be preferred over the auscultatory method as they provide accurate BP measurements and reduce human error (1, 2). For the auscultatory measurement, the clinician inflates the cuff above the expected systolic pressure and gradually releases the air while listening with a stethoscope placed over the brachial artery. As the pressure falls, the pressure at which the first heartbeat is heard is systolic BP. Total disappearance of the sound marks diastolic BP. The same principles are followed to measure BP in a forearm (radial artery) and thigh (popliteal artery). Mechanical devices should be calibrated periodically.

Home blood pressure monitoring, combined with periodic clinician in-office measurements, can improve blood pressure control. However, cuffless home devices (such as smart watches) are not adequately precise, and should not be relied upon (1).

BP is measured in both arms because BP that is > 15 mm Hg higher in one arm than the other requires evaluation of the upper vasculature.

BP is measured in a thigh (with a much larger cuff) to exclude coarctation of the aorta, particularly in patients with diminished or delayed femoral pulses; with coarctation, BP is significantly lower in the legs than in the arms.

Variable measurements and hypertension

White coat hypertension is defined as BP that is elevated in the office but does not meet the criteria for hypertension based on out-of-office readings in untreated patients. The prevalence is approximately 10 to 20% of the population and is more common in children, older adults, and women, as well as those with office-based BP close to the diagnostic threshold.

White coat effect refers to BP that is elevated in the office but is normal during out-of-office readings, among patients who are on treatment for hypertension.

The data about treatment outcomes of patients with white coat hypertension are conflicting. Untreated white coat hypertension seems to be associated with more adverse cardiovascular outcomes compared to normotension. Patients with white coat hypertension have a 3- to 4-fold increased risk of developing sustained hypertension after 7 to 10 years compared to normotensive people (3). White coat effect does not appear to be associated with adverse cardiovascular outcomes.

Masked hypertension is defined as blood pressure that is consistently elevated outside the office, but in which office readings do not meet the criteria for hypertension in untreated people. This group of patients may represent 10 to 30% of the adult population and is more common in men, non-Hispanic Black people, and people with diabetes (4).

Masked hypertension and **masked uncontrolled hypertension** (in those being treated) are associated with increased cardiovascular risk, with mortality similar to the risk in patients with sustained hypertension (5).

Home or ambulatory BP monitoring is indicated when white coat hypertension or masked hypertension is suspected.

History

The history includes the:

- Duration of hypertension and previously recorded BP levels
- History or symptoms of coronary artery disease, heart failure, or obstructive sleep apnea
- Symptoms of, or personal or family history of, other relevant coexisting disorders (eg, stroke, renal dysfunction, peripheral arterial disease, dyslipidemia, diabetes, gout)
- Use of medications that predispose to hypertension (eg, NSAIDs, estrogen-containing oral contraceptives)
- Sleep duration

Social history includes exercise levels and use of tobacco, alcohol, and stimulants (including medications and illicit drugs).

A dietary history focuses on intake of salt and stimulants (eg, tea, coffee, caffeine-containing sodas, energy drinks).

Physical examination

The physical examination includes measurement of height, weight, and waist circumference; fundoscopic examination for retinopathy; auscultation for bruits in the neck and abdomen; and a full cardiac, respiratory, and neurologic examination. The abdomen is palpated for kidney enlargement and abdominal masses. Peripheral arterial pulses are evaluated; diminished or delayed femoral pulses suggest aortic coarctation, particularly in patients < 30 years. A unilateral renal artery bruit may be heard in thin patients with renovascular hypertension.

Testing

After hypertension is diagnosed based on blood pressure measurements, testing is needed to:

- Detect target-organ damage
- Identify cardiovascular risk factors

The more severe the hypertension and the younger the patient, the more extensive is the evaluation. For adults newly diagnosed with hypertension, tests should include (1):

- Urinalysis and urinary albumin:creatinine ratio; if abnormal, consider renal ultrasound
- Lipid panel, complete metabolic panel (including creatinine or cystatin C, potassium, and calcium), fasting plasma glucose or hemoglobin A1c, thyroid-stimulating hormone
- ECG
- Sometimes measurement of plasma free metanephrines (to detect pheochromocytoma)
- Sometimes a sleep study

Depending on results of the examination and initial tests, other tests may be needed. Basic laboratory testing should be repeated at least annually.

Renal ultrasound to evaluate kidney size may provide useful information if urinalysis detects albuminuria (proteinuria), casts, or microhematuria, or if serum creatinine or cystatin C is elevated.

Patients with hypokalemia unrelated to diuretic use are evaluated for high salt intake and for primary aldosteronism by measuring plasma aldosterone levels and plasma renin activity. Screening for primary aldosteronism is also recommended for patients with resistant hypertension, obstructive sleep apnea, or early stroke (< 40 years) (1). The prevalence of primary aldosteronism is higher than previously recognized, occurring in approximately 10 to 20% of patients with resistant hypertension (6, 7).

On ECG, a broad, notched P-wave indicates atrial hypertrophy and, although nonspecific, may be one of the earliest signs of hypertensive heart disease. Elevated QRS voltage, with or without evidence of ischemia, may occur later and indicates left ventricular hypertrophy (LVH). When LVH is seen on ECG, echocardiography is often done.

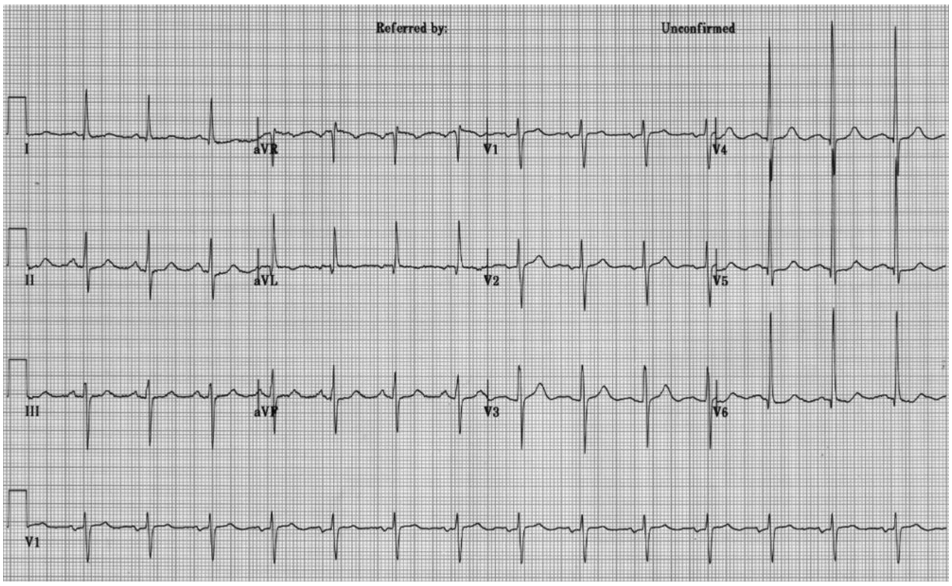
If coarctation of the aorta is suspected, echocardiography, chest CT, or MRI can confirm the diagnosis.

Patients with labile, significantly elevated BP and symptoms such as headache, palpitations, tachycardia, excessive perspiration, tremor, and pallor are screened for pheochromocytoma by measuring plasma free metanephrines and for hyperthyroidism, first by measuring thyroid-stimulating hormone (TSH).

A sleep study should be strongly considered in patients whose history suggests sleep apnea.

Left Ventricular Hypertrophy on ECG

IMAGE



This ECG demonstrates voltage criteria for left ventricular hypertrophy (LVH).

Patients with symptoms suggesting Cushing syndrome, systemic rheumatic diseases, eclampsia, acute porphyria, hyperthyroidism, myxedema, acromegaly, or central nervous system (CNS) disorders also require further evaluation.

Diagnosis references

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Treatment of Hypertension

- Weight loss and exercise
- Smoking cessation
- Adequate sleep duration (a minimum of 6 hours/night)
- Diet: Increased fruits and vegetables, decreased salt, limited alcohol
- Medications: Depending on BP and presence of cardiovascular disease or risk factors

Primary hypertension has no cure, but some causes of secondary hypertension can be corrected. In all cases, control of blood pressure can significantly limit adverse consequences.

Goal blood pressure for most patients is

- BP < 130/80 mm Hg

Goal pressure in part depends on cardiovascular risk, which is best determined by the PREVENT calculator (1). Patients with a 10-year cardiovascular disease (CVD) risk ≥ 7.5% should be encouraged to achieve a systolic BP < 120 mm Hg (2). BP below 130/80 mm Hg appears to continue to reduce the risk of vascular complications. Even older patients, including frail older patients, can tolerate a diastolic BP as low as 60 to 65 mm Hg without an increase in cardiovascular events (3, 4). However, decreasing systolic pressure increases the risk of adverse medication effects (eg, dizziness, light-headedness, possible worsened renal function), and the benefits of lowering BP to levels < 120 mm Hg systolic should be weighed against risks and patient tolerance.

Ideally, patients or family members measure BP at home, provided they have been trained to do so and the sphygmomanometer is regularly calibrated.

Lifestyle modifications

Lifestyle modifications are recommended for all patients with elevated BP or any stage hypertension (2). The best proven nonpharmacologic interventions for prevention and treatment of hypertension include the following:

- Increased physical activity, ideally with a structured exercise program
- Weight loss, if appropriate

- Healthy diet rich in fruits, vegetables, whole grains, and low-fat dairy products, with reduced saturated and total fat content
- Reduced dietary sodium, optimally to < 1500 mg/day (approximately 3.75 g salt) , but at least a 1000 mg/day reduction
- Enhanced dietary potassium intake, unless contraindicated due to chronic kidney disease or use of medications that reduce potassium excretion
- Moderation in alcohol intake in those who drink alcohol to ≤ 2 drinks daily for men and ≤ 1 drink daily for women (one drink is about 12 oz of beer, 5 oz of wine, or 1.5 oz distilled spirits)
- Smoking cessation

Adequate sleep duration (a minimum of 6 hours/night) is also recommended. Short sleep duration (typically defined as < 5 or 6 hours per night in adults, has been associated with hypertension (5). For example, studies suggest that optimizing sleep quality and duration (> 6 hours/night) improves blood pressure control in patients with chronic kidney disease (6).

Dietary modifications can also help control diabetes, obesity, and dyslipidemia. Patients with uncomplicated hypertension do not need to restrict their activities as long as blood pressure is controlled.

Medications

(See also Medications for Hypertension.)

The decision to treat with medication is based on the BP level and the presence of atherosclerotic cardiovascular disease (ASCVD) or its risk factors as determined by the PREVENT calculator (1). Patients with a 10-year predicted risk for CVD events ≥ 7.5% using PREVENT are considered to be at increased risk.

Antihypertensive medication should be initiated for patients with hypertension who meet any of the following criteria (2):

- Average BP ≥ 140/90 mm Hg
- Existing cardiovascular disease (coronary heart disease, stroke, or heart failure) and average BP ≥ 130 mm Hg systolic or ≥ 80 mm Hg diastolic
- Diabetes, chronic kidney disease or 10-year PREVENT cardiovascular risk ≥ 7.5% and average BP ≥ 130 mm Hg systolic or ≥ 80 mm Hg diastolic
- Persistent average BP ≥ 130 mm Hg systolic or ≥ 80 mm Hg diastolic after 3 to 6 months of lifestyle intervention

An important part of management is continued reassessment. If patients are not at goal BP, clinicians should strive to optimize adherence to the current regimen before switching or adding medications.

Treatment of hypertension during pregnancy requires careful medication selection because some antihypertensive medications can harm the fetus.

Medication selection is based on several factors, including comorbidities, contraindications, and tolerability. For most patients, when selecting an agent for **monotherapy,** initial treatment may be with any of the following medication classes:

- Angiotensin-converting enzyme (ACE) inhibitor
- Angiotensin II receptor blocker (ARB)
- Dihydropyridine calcium channel blocker
- Thiazide diuretic (preferably a thiazide-like diuretic such as chlorthalidone or indapamide)

Adults with stage 1 hypertension (systolic BP 130 to 139/diastolic BP 80 to 89 mm Hg) may be treated initially with a single first-line medication, with dose titration and sequential addition of other medications as needed to achieve blood pressure control (2). Adults with stage 2 hypertension (systolic BP ≥ 140/diastolic BP ≥ 90 mm Hg) should be treated initially with 2 first line medications: an ACE inhibitor or ARB combined with either a diuretic or a calcium channel blocker. When combination therapy is indicated, starting treatment with single pill combinations improves adherence (7, 8).

Signs of hypertensive emergencies require immediate blood pressure reduction with parenteral antihypertensives. Such signs may include seizures or focal neurologic symptoms, retinal hemorrhage, chest pain, dyspnea,or nausea and vomiting.

Some antihypertensives are avoided in certain disorders (eg, ACE inhibitors in severe aortic stenosis) whereas others are preferred for certain disorders (eg, calcium channel blockers for patients with angina pectoris, ACE inhibitors or ARBs for patients with diabetes and proteinuria—see tables Initial Choice of Antihypertensive Medication Class and Antihypertensives for Patients With Comorbidities).

If the goal BP is not achieved within 1 month, assess adherence and tolerability and reinforce the importance of following treatment. If patients *are* adherent with the regimen, the dose of the initial medication can be increased or a second medication added (selected from among medications recommended for initial treatment). Note that an ACE inhibitor and an ARB should not be used together. Therapy is titrated frequently.

If target BP cannot be achieved with 2 medications, a third medication from choices of initial medications is added. If such a third medication is not tolerated or is contraindicated, a medication from another class (eg, aldosterone antagonist) can be used. Patients with such difficult to control BP may benefit from consultation with a hypertension specialist. For resistant hypertension (BP remains above goal despite use of 3 different antihypertensive medications), 4 or more medications are commonly needed.

TABLE

Initial Choice of Antihypertensive Medication Class

Medication Class	Indications
ACE inhibitors*	Youth

Medication Class	Indications
	Left ventricular failure due to systolic dysfunction† Diabetes with nephropathy Albumin:creatinine ratio > 30 mg/gram (> 33.9 mg/mmol) in patients with chronic renal disorders or diabetic glomerulosclerosis Erectile dysfunction due to other medications
Angiotensin II receptor blockers*	Youth Conditions for which ACE inhibitors are indicated but not tolerated because of cough or angioedema Diabetes with nephropathy Left ventricular failure with systolic dysfunction Secondary stroke
Long-acting calcium channel blockers	Older age Angina pectoris Arrhythmias (eg, atrial fibrillation, paroxysmal supraventricular tachycardia) Isolated systolic hypertension in older patients (dihydropyridines)† High CAD risk (nondihydropyridines)†
Thiazide-type diuretics† (chlorthalidone or indapamide)	Older age Heart failure

* Contraindicated in pregnancy.

† Reduced morbidity and mortality rates in randomized studies.

ACE = angiotensin-converting enzyme; CAD = coronary artery disease.

Achieving adequate blood pressure control often requires several evaluations and changes in pharmacotherapy. Reluctance to titrate or add medications to control BP must be overcome. Nonadherence to therapy, particularly because lifelong treatment is required, can interfere with adequate BP control. Education, with empathy and support, is essential for success.

TABLE

Antihypertensives for Patients With Comorbidities

Comorbidity	Medication Classes
Coronary artery disease or risk factors	Beta-blockers ACE inhibitors Diuretics Calcium channel blockers
Chronic kidney disorders	ACE inhibitors Angiotensin II receptor blockers
Diabetes	Diuretics ACE inhibitors Angiotensin II receptor blockers Calcium channel blockers
Heart failure	ACE inhibitors Angiotensin II receptor blockers Beta-blockers Potassium-sparing diuretics Other diuretics*
Post-myocardial infarction	ACE inhibitors Beta-blockers Spironolactone or eplerenone
Risk of recurrent stroke	ACE inhibitors Angiotensin II receptor blockers Calcium channel blockers Diuretics

* Long-term diuretic use may increase mortality in patients with heart failure who do not have pulmonary congestion.

ACE = angiotensin-converting enzyme.

Devices and physical interventions

Renal artery sympathetic nerve ablation is a potential option for uncontrolled hypertension, using percutaneous catheter-based radiofrequency devices, ultrasound, or the injection of neurolytic agents into tissues surrounding the renal vasculature (9). Several industry-funded sham-controlled studies with different patient populations (eg, untreated hypertension [10], treated hypertension [11], or resistant hypertension [12]) have demonstrated statistically and/or clinically significant reductions in systolic blood pressure. However, whether these devices reduce major cardiovascular events remains uncertain, with studies ongoing related to efficacy, patient selection, safety, and long-term impact on quality of life (13).

Baroreflex activation therapy has been proposed for management of resistant hypertension, but studies have not provided sufficient evidence to recommend the use of these devices (2). The therapy employs a battery-powered device surgically implanted around the carotid body to stimulate the baroreceptor and, in a dose-dependent manner, lower blood pressure. In one long-term follow-up study of patients with resistant hypertension who were included in earlier pivotal trials, baroreflex activation therapy maintained its efficacy for persistent reduction of office BP without major safety issues (14).

Resistant hypertension

Resistant hypertension is defined as blood pressure that is not controlled to goal despite adherence to 3 suitably dosed antihypertensive medications of different classes (including a diuretic) and after white coat effect is excluded. Blood pressure that requires 4 drugs to achieve control is also considered controlled resistant hypertension. Use of a mineralocorticoid receptor antagonist can be beneficial in attaining goal blood pressure (15).

Secondary causes of hypertension must be excluded as part of the evaluation for resistant hypertension. Hyperaldosteronism or subclinical hypercortisolism has been identified in 15 to 20% of cases of resistant hypertension.

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Prognosis for Hypertension

The higher the blood pressure, and the more severe the retinal changes and other evidence of target-organ involvement, the worse the prognosis. Systolic BP predicts fatal and nonfatal cardiovascular events better than diastolic BP (1, 2). Effective control of hypertension prevents most complications and prolongs life.

Without treatment, 1-year survival is < 10% in patients with retinal sclerosis, cotton-wool exudates, arteriolar narrowing, and hemorrhage (grade 3 retinopathy) and is < 5% in patients with the same changes plus papilledema (grade 4 retinopathy [3]).

Coronary artery disease is the most common cause of death among treated patients. Ischemic or hemorrhagic stroke is a common consequence of inadequately treated hypertension.

Prognosis references

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Key Points

- **Only about 50% of patients in the United States with hypertension receive treatment, and about one-quarter of those patients have adequate blood pressure (BP) control.**
- **Most hypertension is primary; only 5 to 15% is secondary to another disorder (eg, primary aldosteronism, renal parenchymal disease).**
- **Severe or prolonged hypertension damages the cardiovascular system, brain, and kidneys, increasing risk of myocardial infarction, stroke, and chronic kidney disease.**
- **Hypertension is usually asymptomatic until complications develop in target organs.**
- **When hypertension is newly diagnosed, do a urinalysis, spot urine albumin:creatinine ratio, blood tests (creatinine, potassium, sodium, calcium, fasting plasma glucose, lipid panel, and thyroid-stimulating hormone), and ECG.**
- **Determine the 10-year cardiovascular risk, using the PREVENT calculator.**
- **Reduce BP to < 130/80 mm Hg for most patients..**

- Patients, including those with a kidney disorder or diabetes, whose 10-year CVD risk is ≥ 7.5% should be encouraged to achieve a systolic BP < 120 mm Hg. A higher target might be appropriate for frail older patients who become symptomatic at lower pressures.
- Treatment involves lifestyle changes, especially a low-sodium and higher potassium diet, medication (including thiazide diuretics, angiotensin-converting enzyme inhibitors, angiotensin II receptor blockers, and dihydropyridine calcium channel blockers), and management of secondary causes of hypertension if present.

Drug Information for the Topic

MERCK MANUAL
Professional Version

Hypertensive Emergencies

(Acute Severe Hypertension)

Full Review: Feb 2025 By ***Matthew R. Weir****, MD, University of Maryland School of Medicine | Peer reviewed by* ***Jonathan G. Howlett****, MD, Cumming School of Medicine, University of Calgary*

Last updated: Apr 2026

A hypertensive emergency is severe hypertension (often defined as systolic blood pressure (BP) ≥ 180 mm Hg and/or diastolic blood pressure ≥ 120 mm Hg) with signs of damage to target organs (primarily the brain, cardiovascular system, and kidneys). Diagnosis is by BP measurement, ECG, urinalysis, and serum electrolyte and creatinine measurements. Treatment is immediate BP reduction with IV antihypertensives.

(See also Hypertension.)

Signs of target-organ damage include (1):

- Hypertensive encephalopathy
- Hypertensive retinopathy
- Preeclampsia or eclampsia
- Acute left ventricular failure with pulmonary edema
- Myocardial ischemia
- Acute aortic dissection
- Renal failure

Damage is rapidly progressive and sometimes fatal.

Hypertensive encephalopathy may involve a failure of cerebral autoregulation of blood flow. Normally, as blood pressure increases, cerebral vessels constrict to maintain constant cerebral perfusion. Above a mean arterial pressure (MAP) of about 160 mm Hg (lower for people who are normotensive whose BP suddenly increases), the cerebral vessels begin to dilate rather than remain constricted. As a result, the very high BP is transmitted directly to the capillary bed with transudation and exudation of plasma into the brain, causing cerebral edema, including papilledema.

Although many patients with stroke and intracranial hemorrhage present with elevated BP, elevated BP is often a consequence rather than a cause of the condition. Whether rapidly lowering BP is beneficial in these conditions is unclear (2); it may even be harmful.

CLINICAL CALCULATORS

Mean Vascular Pressure (systemic or pulmonary)

Hypertensive urgencies

Severe hypertension (eg, systolic pressure > 180 mm Hg) without target-organ damage (except perhaps grades 1 to 2 retinopathy) may be considered a hypertensive urgency. Although BP at these very high levels often concerns clinicians, acute complications are unlikely, so immediate BP reduction is not required.

Anxiety is by far the most common cause of hypertensive urgency. Treating the anxiety by counseling and/or use of anxiety-alleviating medications is often helpful in reducing BP (3) and thus reducing the incidence of hypertensive urgencies. If elevated BP persists, patients should be started on medications, such as combination therapy with 2 oral antihypertensives (for severe hypertension), and closely evaluated for treatment efficacy on an outpatient basis.

General references

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Symptoms and Signs of Hypertensive Emergencies

Blood pressure is elevated, often markedly (systolic pressure > 180 mm Hg and/or diastolic pressure ≥ 120 mm Hg). Central nervous system symptoms include rapidly changing neurologic abnormalities (eg, confusion, transient cortical blindness, hemiparesis, hemisensory defects, seizures). Cardiovascular symptoms include chest pain and dyspnea. Renal involvement may be asymptomatic, although severe azotemia due to advanced renal failure may cause lethargy or nausea.

Physical examination focuses on target organs, with neurologic examination, funduscopy, and cardiovascular examination. Global cerebral deficits (eg, confusion, obtundation, coma), with or without focal deficits, suggest encephalopathy; normal mental status with focal deficits suggests stroke.

Severe retinopathy (sclerosis, cotton-wool spots, arteriolar narrowing, hemorrhage, papilledema) is usually present with hypertensive encephalopathy, and some degree of retinopathy is present in many other hypertensive emergencies.

Jugular venous distention, basilar lung crackles, and a third heart sound suggest pulmonary edema.

Asymmetry of pulses between arms suggests aortic dissection.

Diagnosis of Hypertensive Emergencies

- Systolic blood pressure > 180 mm Hg
- Testing to identify target-organ involvement: ECG, urinalysis, serum electrolytes, and creatinine; if neurologic findings, head CT

Testing typically includes ECG, urinalysis, serum electrolytes, and creatinine.

Patients with neurologic findings require head CT to diagnose intracranial bleeding, edema, or infarction.

Patients with chest pain or dyspnea require an ECG and chest radiograph. ECG abnormalities suggesting acute target-organ damage include acute ischemic changes.

Urinalysis abnormalities typical of renal involvement include red blood cells (RBCs), RBC casts, and proteinuria.

Diagnosis is based on the presence of a BP > 180 mm Hg and findings of target-organ involvement.

Treatment of Hypertensive Emergencies

- Initiate short-acting IV medication (eg, labetalol, clevidipine, esmolol) in the emergency department
- Admit to intensive care unit (ICU) where titratable IV medications (eg, nitroprusside, fenoldopam, nicardipine) can be monitored
- Goal: 20 to 25% reduction MAP in 1 to 2 hours

Hypertensive emergencies are treated in an ICU; blood pressure is progressively (although not abruptly) reduced using a short-acting, titratable IV medication. Choice of medication and speed and degree of reduction vary somewhat with the target organ involved, but generally a 20 to 25% reduction in MAP over about an hour or two is appropriate, with further titration based on symptoms. Achieving "normal" BP urgently is not necessary. Typical first-line medications include nitroprusside, fenoldopam, nicardipine, and labetalol (see table Parenteral Medications for Hypertensive Emergencies). Nitroglycerin alone is less potent.

TABLE

Parenteral Medications for Hypertensive Emergencies

Medication	Selected Adverse Effects*	Special Indications
Clevidipine	Atrial fibrillation, fever, insomnia, nausea, headache	Most hypertensive emergencies Should be used cautiously in patients with acute heart failure
Enalaprilat	Precipitous fall in blood pressure in high-renin states, variable response	Acute left ventricular failure Should be avoided in acute myocardial infarction
Esmolol	Hypotension, nausea	Aortic dissection perioperatively
Fenoldopam	Tachycardia, headache, nausea, flushing, hypokalemia, elevation of intraocular pressure in patients with glaucoma	Most hypertensive emergencies Should be used cautiously in patients with myocardial ischemia
Hydralazine	Tachycardia, flushing, headache, vomiting, aggravation of angina	Eclampsia
Labetalol	Vomiting, scalp tingling, burning in throat, dizziness, nausea, heart block, orthostatic hypotension	Most hypertensive emergencies, except acute left ventricular failure Should be avoided in patients with asthma
Nicardipine	Tachycardia, headache, flushing, local phlebitis	Most hypertensive emergencies, except acute heart failure Should be used cautiously in patients with myocardial ischemia
Nitroglycerin	Headache, tachycardia, nausea, vomiting,	Myocardial ischemia, heart failure

Medication	Selected Adverse Effects*	Special Indications
	apprehension, restlessness, muscular twitching, palpitations, methemoglobinemia, tolerance with prolonged use	
Nitroprusside	Nausea, vomiting, agitation, muscle twitching, sweating, cutis anserina (if blood pressure is reduced too rapidly), thiocyanate and cyanide toxicity	Most hypertensive emergencies Should be used cautiously in patients with high intracranial pressure or azotemia
Phentolamine	Tachycardia, flushing, headache	Rarely used unless patient has a confirmed diagnosis of pheochromocytoma

* Hypotension may occur with all medications.

† A special delivery system (eg, infusion pump for nitroprusside, nonpolyvinyl chloride tubing for nitroglycerin) is required.

Oral medications are not indicated because onset is variable and the medications are difficult to titrate. Although short-acting oral nifedipine reduces blood pressure rapidly, it may lead to acute hypotension, which may lead to cardiovascular and cerebrovascular ischemic events (sometimes fatal) and is therefore not recommended.

Clevidipine is an ultra-short-acting (within 1 to 2 minutes), third-generation calcium channel blocker that reduces peripheral resistance without affecting venous vascular tone and cardiac filling pressures. Clevidipine is rapidly hydrolyzed by blood esterases and, thus, its metabolism is not affected by renal or hepatic function. It has been shown to be effective and safe in the control of perioperative hypertension and hypertensive emergencies and was associated with lower mortality than nitroprusside (1). Clevidipine may thus be preferred over nitroprusside for most hypertensive emergencies, although it should be used with caution in acute heart failure with reduced ejection fraction as it may have negative inotropic effects. If clevidipine is not available, then fenoldopam, nitroglycerin, or nicardipine are reasonable alternatives.

Fenoldopam is a peripheral dopamine-1 agonist that causes systemic and renal vasodilation and natriuresis. Onset is rapid and half-life is brief, making it an effective alternative to nitroprusside, with the added benefit that it does not cross the blood-brain barrier.

Labetalol is a beta-blocker with some alpha-1-blocking effects, thus causing vasodilation without the typical accompanying reflex tachycardia. It can be given as a constant infusion or as frequent boluses; use of boluses has not been shown to cause significant hypotension. Labetalol is used during pregnancy, for intracranial disorders requiring BP control, and after myocardial infarction. Adverse effects are minimal, but because of its beta -blocking activity, labetalol should not be used for hypertensive emergencies in patients with asthma. Low doses may be used for left ventricular failure if nitroglycerin is given simultaneously.

Nitroprusside is a venous and arterial dilator, reducing preload and afterload. Data show increased mortality with nitroprusside compared to clevidipine, nitroglycerin, and nicardipine, and thus nitroprusside should not be used when other alternatives are available. Its use is mostly limited to patients with acute decompensated heart failure who are hypertensive. It is also used for hypertensive encephalopathy and, with beta-blockers, for aortic dissection. The medication is rapidly broken down into cyanide and nitric oxide (the active moiety). Cyanide is detoxified to thiocyanate. However, administration of > 2 mcg/kg/minute can lead to cyanide accumulation with toxicity to the central nervous system and heart; manifestations include agitation, seizures, cardiac instability, and an anion gap metabolic acidosis.

Prolonged administration of nitroprusside (> 1 week or, in patients with renal insufficiency, 3 to 6 days) leads to accumulation of thiocyanate, which can result in lethargy, tremor, abdominal pain, and vomiting. Other adverse effects include transitory elevation of hair follicles (cutis anserina) if BP is reduced too rapidly. Thiocyanate levels should be monitored daily after 3 consecutive days of therapy, and the medication should be stopped if the serum thiocyanate level is > 12 mg/dL (> 2 mmol/L). Because nitroprusside is broken down by ultraviolet light, the IV bag and tubing are wrapped in an opaque covering.

Nitroglycerin is a vasodilator that affects veins more than arterioles. It can be used to manage hypertension during and after coronary artery bypass graft surgery, acute myocardial infarction, unstable angina pectoris, and acute pulmonary edema. IV nitroglycerin is preferable to nitroprusside for patients with severe coronary artery disease because nitroglycerin increases coronary flow, whereas nitroprusside tends to decrease coronary flow to ischemic areas, possibly because of a “steal” mechanism.

For long-term BP control, nitroglycerin must be used with other medications. The most common adverse effect is headache, occurring in the majority of patients (2); others include tachycardia, nausea, vomiting, apprehension, restlessness, muscular twitching, and palpitations.

Nicardipine, a dihydropyridine calcium channel blocker with less negative inotropic effects than nifedipine, acts primarily as a vasodilator. It is most often used for postoperative hypertension and during pregnancy. It may cause flushing, headache, and tachycardia; it can decrease glomerular filtration rate (GFR) in patients with renal insufficiency.

Treatment references

CLINICAL CALCULATORS

Mean Vascular Pressure (systemic or pulmonary)

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Key Points

- A hypertensive emergency is significantly elevated blood pressure (eg, systolic blood pressure > 180 mm Hg and/or diastolic pressure ≥ 120 mm Hg) that causes target-organ damage; it requires intravenous therapy and hospitalization.
- Target-organ damage includes hypertensive encephalopathy, preeclampsia and eclampsia, acute left ventricular failure with pulmonary edema, myocardial ischemia, acute aortic dissection, and renal failure.
- Do ECG and urinalysis, measure serum electrolytes and creatinine, and, for patients with neurologic symptoms or signs, obtain head CT.
- Reduce mean arterial pressure by about 20 to 25% over the first hour using a short-acting, titratable IV medication such as clevidipine, nitroglycerin, fenoldopam, nicardipine, or labetalol.
- It is not necessary to achieve "normal" blood pressure urgently (especially true in acute stroke).

Drug Information for the Topic

MERCK MANUAL
Professional Version

Renovascular Hypertension

*Full Review: Feb 2025 By **Matthew R. Weir**, MD, University of Maryland School of Medicine | Peer reviewed by **Jonathan G. Howlett**, MD, Cumming School of Medicine, University of Calgary*
Last updated: Dec 2025

Renovascular hypertension is blood pressure elevation due to partial or complete occlusion of one or more renal arteries or their branches. It is usually asymptomatic unless long-standing. A bruit can be heard over one or both renal arteries in < 50% of patients. Diagnosis is by physical examination and renal imaging with duplex ultrasound, CT angiography, or magnetic resonance angiography. Treatment is with medications and sometimes revascularization by percutaneous angioplasty or surgery.

(See also Hypertension.)

Renovascular disease is one of the most common causes of reversible hypertension but accounts for < 1% of all cases of hypertension (1). Stenosis or occlusion of a main renal artery, an accessory renal artery, or any of their branches can cause hypertension by stimulating release of renin from juxtaglomerular cells of the affected kidney. The area of the arterial lumen must be decreased by ≥ 70% and a significant poststenotic gradient must be present before stenosis is likely to contribute to blood pressure (BP) elevation.

Overall, about 80% of cases are caused by renal atherosclerosis and 20% by fibromuscular dysplasia. Atherosclerosis is more common among men > 50 years and affects mainly the proximal one-third of the renal artery. Fibromuscular dysplasia is more common among younger patients (usually women) and usually affects the distal two-thirds of the main renal artery and the branches of the renal arteries. Rarer causes include emboli, trauma, inadvertent ligation during surgery, and extrinsic compression of the renal pedicle by tumors.

Renovascular hypertension is characterized by high cardiac output and high peripheral resistance.

General reference

1. Dworkin LD, Cooper CJ. Clinical practice. Renal-artery stenosis. *N Engl J Med.* 2009;361(20):1972-1978. doi:10.1056/NEJMcp0809200

Symptoms and Signs of Renovascular Hypertension

Renovascular hypertension is usually asymptomatic. A systolic-diastolic bruit in the epigastrium, usually transmitted to one or both upper quadrants and sometimes to the back, is almost pathognomonic, but it is present in only about 50% of patients with fibromuscular dysplasia and is rare in patients with renal atherosclerosis.

Renovascular hypertension should be suspected if:

- Diastolic hypertension develops abruptly
- New or previously stable hypertension rapidly worsens over a period of 6 months
- Hypertension is initially severe (eg, systolic BP > 180 mm Hg)
- There is unexplained worsening renal function
- Hypertension is resistant to medications

A history of trauma to the back or flank or acute pain in this region with or without hematuria suggests renovascular hypertension (possibly due to arterial injury), but these findings are rare.

Asymmetric renal size (> 1 cm difference) discovered incidentally during imaging tests, and recurrent episodes of unexplained acute pulmonary edema or heart failure also suggest renovascular hypertension.

Diagnosis of Renovascular Hypertension

- Initial identification with ultrasound, magnetic resonance angiography, or CT angiography
- Confirmation with renal angiography (also may be therapeutic)

If renovascular hypertension is suspected, ultrasound, magnetic resonance angiography, or CT angiography are reasonable non-invasive initial tests. Noninvasive testing is less reliable for detecting fibromuscular dysplasia because of the location of the stenotic lesion (which typically involves the intrarenal portion of the renal artery).

Renal angiography is the definitive test, which may be done when noninvasive testing is inconclusive and the clinical suspicion is high.

Duplex Doppler ultrasound can assess renal blood flow and is a reliable, noninvasive method for identifying significant stenosis (eg, > 60%) in the main renal arteries. Sensitivity and specificity are 85 to 90% when experienced technicians do the test (1). It is less accurate in patients with branch stenosis.

Magnetic resonance angiography is a more accurate and specific noninvasive test to assess the renal arteries (2). However, concerns about gadolinium-associated complications, including nephrogenic systemic fibrosis, limit its use, particularly among patients with renal insufficiency.

Renovascular Hypertension (Magnetic Resonance Angiography)

IMAGE

Magnetic resonance angiography demonstrates severe stenosis at the origin of the left main renal artery.

IMAGE PROVIDED BY JAN N. BASILE, MD.

CT angiography is another noninvasive test with sensitivities and specificities in the range of 95% (3). Its use is also limited by exposure to radiocontrast media in patients with renal insufficiency.

Renal angiography is done if noninvasive imaging methods indicate a lesion amenable to angioplasty or stenting or if the results of other screening tests are inconclusive. Digital subtraction angiography with selective injection of the renal arteries can also confirm the diagnosis, but angioplasty or stent placement cannot be done in the same procedure.

Radionuclide imaging is rarely used for diagnostic purposes and is occasionally used as a functional test to compare blood flow and filtration between the kidneys. Imaging is performed before and after an oral dose of captopril. The angiotensin-converting enzyme (ACE) inhibitor causes the affected artery to narrow, decreasing perfusion on the scintiscan. Narrowing also causes an increase in serum renin, which is measured before and after captopril administration. This test may be less reliable in patients with decreased renal function.

Renal Artery Stenosis (Digital Subtraction Angiography)

IMAGE

The left image shows 90% stenosis at the origin of the left main renal artery on digital subtraction angiography. The right image shows complete resolution of stenosis after stent placement.

IMAGE PROVIDED BY JAN N. BASILE, MD.

Measurements of renal vein renin activity are sometimes misleading and, unless surgery is being considered, are not necessary. However, in unilateral disease, a renal vein renin activity ratio of > 1.5 (affected to unaffected side) usually predicts a good outcome with revascularization. The test is done when patients are depleted of sodium, stimulating the release of renin.

Diagnosis references

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Treatment of Renovascular Hypertension

- Aggressive medical management of hypertension, atherosclerosis, and related disorders
- For fibromuscular dysplasia, sometimes angioplasty with or without stent placement
- Rarely bypass grafting

Without treatment, the prognosis is similar to that for patients with untreated primary hypertension.

All patients should have aggressive medical management of their hypertension.

Atherosclerotic renal artery stenosis

For patients with atherosclerotic renal artery stenosis, data from a large, randomized trial (the cardiovascular outcomes in renal atherosclerotic lesions [CORAL] trial) showed that stent placement did not improve outcomes compared to medical management alone (1). Although stent placement did provide a small (-2 mm Hg), statistically significant decrease in systolic blood pressure, there was no significant clinical benefit for prevention of stroke, myocardial infarction, heart failure, death due to cardiovascular or renal disease, or progression of kidney disease (including the need for renal replacement therapy). Importantly, all patients in the CORAL study received aggressive medical management of their hypertension and diabetes (if present), along with antiplatelet medications and a statin to manage atherosclerosis.

Thus, the decision to eschew angioplasty must be accompanied by strict adherence to current medical management guidelines (2). If serum creatinine increases by > 50% with optimal medical treatment for blood pressure, renal artery stenting may help preserve kidney function (3). For patients unable to strictly adhere to medical management guidelines and with a > 70% renal artery stenosis, stent placement may still be considered.

Fibromuscular dysplasia

For most patients with fibromuscular dysplasia of the renal artery, percutaneous transluminal angioplasty (PTA) is recommended. Placement of a stent reduces the risk of restenosis; antiplatelet medications (eg, aspirin, clopidogrel) are given afterward. Saphenous vein bypass grafting is recommended only when extensive disease in the renal artery branches makes PTA technically unfeasible. Sometimes complete surgical revascularization requires microvascular techniques that can only be done ex vivo with autotransplantation of the kidney. Cure rate is 90% in appropriately selected patients; surgical mortality rate is < 1% (4).

Medical treatment is always preferable to nephrectomy in patients whose kidneys cannot be revascularized for technical reasons.

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Key Points

- **Stenosis (> 70%) or occlusion of a renal artery can cause hypertension by stimulating release of renin from juxtaglomerular cells of the affected kidney.**
- **About 80% of cases are caused by atherosclerosis, and 20% by fibromuscular dysplasia.**
- **Suspect a renovascular cause if diastolic hypertension develops abruptly; if new or previously stable hypertension rapidly worsens within 6 months; or if hypertension is initially severe (systolic > 180 mm Hg), associated with worsening renal function, or highly refractory to treatment with medication.**
- **Do ultrasound, magnetic resonance angiography, or CT angiography to identify patients who should have renal angiography, the definitive test.**
- **Give aggressive medical treatment of hypertension, atherosclerosis, and related disorders.**
- **For patients with fibromuscular dysplasia, consider percutaneous transluminal angioplasty and/or stent placement or rarely a vascular bypass graft.**

Drug Information for the Topic

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Professional Version

Medications for Hypertension

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Last updated: Dec 2025

The treatment of hypertension involves lifestyle modifications (eg, dietary modification, weight loss, exercise), alone or in combination with medications. The decision to treat with medication is based on the blood pressure (BP) level, the presence of atherosclerotic cardiovascular disease (ASCVD) or its risk factors, and other considerations.

A number of medication classes are effective for initial and subsequent management of hypertension (1):

- Diuretics
- Angiotensin-converting enzyme (ACE) inhibitors
- Angiotensin II receptor blockers (ARBs)
- Calcium channel blockers
- Direct renin inhibitors
- Beta-blockers
- Alpha adrenergic modifiers
- Direct vasodilators

Selection and use of medications in the treatment of stable hypertension is discussed elsewhere. For treatment of hypertensive emergencies, see table Parenteral Medications for Hypertensive Emergencies.

(See also Hypertension and Hypertensive Emergencies.)

General medication reference

1. Writing Committee Members*, Jones DW, Ferdinand KC, et al. 2025 AHA/ACC/AANP/AAPA/ABC/ACCP/ACPM/AGS/AMA/ASPC/NMA/PCNA/SGIM Guideline for the Prevention, Detection, Evaluation and Management of High Blood Pressure in Adults: A Report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Hypertension.* 2025;82(10):e212-e316. doi:10.1161/HYP.0000000000000249

Diuretics for Hypertension

Main classes of diuretics used for hypertension (see table Oral Diuretics for Hypertension) are

- Thiazide diuretics
- Potassium-sparing diuretics
- Loop diuretics

Diuretics modestly reduce plasma volume and reduce vascular resistance, possibly via shifts in sodium from intracellular to extracellular loci.

Thiazide-type diuretics (thiazide diuretics and thiazide-like diuretics) are most commonly used. Chlorthalidone and indapamide, thiazide-like diuretics, are preferred over hydrochlorothiazide because of their higher potency (1) and longer duration of action. However, one retrospective analysis from The Diuretic Comparison Project (2), and another pragmatic trial conducted in Veterans (3) demonstrated that chlorthalidone, despite its greater potency, did not have an advantage over hydrochlorothiazide in reducing cardiovascular or kidney disease progression events. Although thiazide diuretics had been thought to be ineffective in patients with stage 4 chronic kidney disease, chlorthalidone has been shown to be effective in improving blood pressure in patients with glomerular filtration rates < 30 mL/minute (4). In addition to other antihypertensive effects, thiazide diuretics cause a small amount of vasodilation when intravascular volume is normal. In a few patients, thiazide-type diuretics can increase serum cholesterol slightly (mostly low-density lipoprotein cholesterol) and also increase triglyceride levels, although these effects may not persist > 1 year (5). The increase is apparent within 4 weeks of treatment and can be ameliorated by a low-fat diet. The possibility of a slight increase in lipid levels does not contraindicate diuretic use in patients with dyslipidemia.

Potassium-sparing diuretics are not as effective as thiazide-type diuretics in controlling hypertension and thus are not used for initial treatment. They do not cause hypokalemia, hyperuricemia, or hyperglycemia, and thus may be helpful for patients who develop these adverse effects of thiazide diuretics. Potassium-sparing diuretics or potassium supplements are not needed when an angiotensin-converting enzyme (ACE) inhibitor or angiotensin II receptor blocker is used because these medications increase serum potassium.

Loop diuretics are used to treat hypertension only in patients who have an estimated glomerular filtration rate (GFR) < 30 mL/minute; these diuretics are given at least twice a day (except for torsemide which can be given once a day).

All diuretics except the potassium-sparing distal tubular diuretics (eg, spironolactone) cause significant potassium loss, so serum potassium is measured monthly until the level stabilizes. Unless serum potassium is normalized, potassium channels in the arterial walls close and the resulting vasoconstriction makes achieving the blood pressure (BP) goal difficult. Patients with potassium levels < 3.5 mEq/L (< 3.5 mmol/L) are given potassium supplements. Supplements may be continued long-term at a lower dose, or a potassium-sparing diuretic (eg, spironolactone, triamterene, amiloride) may be added. Addition of a potassium-sparing diuretic or potassium supplements is also recommended for patients who are also taking digoxin, have a known heart disorder, have an abnormal ECG, have ectopy or arrhythmias, or develop ectopy or arrhythmias while taking a diuretic.

In most patients with diabetes, thiazide-type diuretics do not affect control of diabetes. Uncommonly, diuretics precipitate or worsen type 2 diabetes in patients with metabolic syndrome.

A hereditary predisposition probably explains the few cases of gout due to diuretic-induced hyperuricemia. Diuretic-induced hyperuricemia without gout does not require treatment or discontinuation of the diuretic.

Diuretics may slightly increase mortality in patients with a history of heart failure who do not have pulmonary congestion, particularly in those who are also taking an ACE inhibitor or angiotensin II receptor blocker and who do not drink at least 1400 mL (48 oz) of fluid daily. The increased mortality is probably related to diuretic-induced hyponatremia and hypotension.

TABLE

Oral Diuretics for Hypertension

Medication	Selected Adverse Effects
Thiazides and thiazide-like diuretics (chlorthalidone and indapamide)	
Chlorothiazide	Hypokalemia (which increases digitalis toxicity), hyperuricemia, glucose intolerance, hypercholesterolemia, hypertriglyceridemia, hypercalcemia, sexual dysfunction in men, weakness, rash
Chlorothiazide	
Hydrochlorothiazide	
Indapamide	
Potassium-sparing diuretics	
Amiloride	Hyperkalemia (particularly in patients with renal failure and in patients treated with an angiotensin-converting enzyme inhibitor, angiotensin II receptor blocker, or nonsteroidal anti-inflammatory drug), nausea, gastrointestinal distress, gynecomastia, menstrual irregularities (with spironolactone)
Eplerenone*	
Spironolactone*	
Triamterene	
Loop diuretics	
Bumetanide	Hypokalemia, hyponatremia, hypomagnesemia, dehydration, postural hypotension, tinnitus, hearing loss
Ethacrynic acid	
Furosemide	
Torsemide	

* Aldosterone receptor blockers.

Diuretic references

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2. Ishani A, Cushman WC, Leatherman SM, et al. Chlorthalidone vs. Hydrochlorothiazide for Hypertension-Cardiovascular Events. *N Engl J Med.* 2022;387(26):2401-2410. doi:10.1056/NEJMoa2212270
3. Ishani A, Hau C, Raju S, et al. Chlorthalidone vs Hydrochlorothiazide and Kidney Outcomes in Patients With Hypertension: A Secondary Analysis of a Randomized Clinical Trial. *JAMA Netw Open.* 2024;7(12):e2449576. doi:10.1001/jamanetworkopen.2024.49576

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5. Ott SM, LaCroix AZ, Ichikawa LE, Scholes D, Barlow WE. Effect of low-dose thiazide diuretics on plasma lipids: results from a double-blind, randomized clinical trial in older men and women. *J Am Geriatr Soc.* 2003;51(3):340-347. doi:10.1046/j.1532-5415.2003.51107.x

ACE Inhibitors for Hypertension

ACE inhibitors (see table Oral ACE Inhibitors and Angiotensin II Receptor Blockers for Hypertension) reduce blood pressure by interfering with the conversion of angiotensin I to angiotensin II and by inhibiting the degradation of bradykinin, thereby decreasing peripheral vascular resistance without causing reflex tachycardia. These medications reduce BP in many patients with hypertension, regardless of plasma renin activity. Because these medications provide renal protection, they (along with angiotensin II receptor blockers) are the medications of choice for patients with diabetes. They are not recommended for initial treatment in patients with African ancestry, in whom they appear to increase the risk of stroke when used for initial treatment (1).

A dry, irritating cough is the most common adverse effect, with estimates of occurrence in up to 20% in North American and Europe populations and up to 40% in Asian populations (2, 3). Angioedema is the most serious adverse effect and, if it affects the oropharynx, can be fatal. Angioedema is most common among patients with African ancestry and those who smoke.

ACE inhibitors may increase serum potassium and creatinine levels, especially in patients with chronic kidney disease and those taking potassium-sparing diuretics, potassium supplements, or nonsteroidal anti-inflammatory drugs (NSAIDs).

ACE inhibitors are contraindicated during pregnancy.

In patients with a renal disorder, serum creatinine and potassium levels are monitored at least every 3 months. Patients who have stage 3 nephropathy (estimated GFR of < 60 mL/minute to > 30 mL/minute) and are given ACE inhibitors can usually tolerate an increase in serum creatinine up to 30 to 35% above baseline. ACE inhibitors can cause acute kidney injury in patients who have hypovolemia, severe heart failure, severe bilateral renal artery stenosis, or severe stenosis in the artery to a solitary kidney.

Thiazide-type diuretics enhance the antihypertensive activity of ACE inhibitors and angiotensin II receptor blockers more than that of other classes of antihypertensives (4, 5). Spironolactone and eplerenone also appear to enhance the effect of ACE inhibitors.

TABLE

Oral Angiotensin-Converting Enzyme (ACE) Inhibitors and Angiotensin II Receptor Blockers (ARBs) for Hypertension

Medication	Selected Adverse Effects

Medication	Selected Adverse Effects
ACE inhibitors*	
Benazepril	Rash, cough, angioedema, hyperkalemia (particularly in patients with renal insufficiency or taking nonsteroidal anti-inflammatory drugs, potassium-sparing diuretics, or potassium supplements), dysgeusia, reversible acute kidney injury if stenosis affecting one or both kidneys threatens renal function, proteinuria (rare at recommended doses), neutropenia (rare), hypotension with initiation of treatment (particularly in patients with high plasma renin activity or with hypovolemia due to diuretics or other conditions)
Captopril	
Enalapril	
Fosinopril	
Lisinopril	
Perindopril	
Quinapril	
Ramipril	
Trandolapril	
ARBs	
Azilsartan	Dizziness, angioedema (very rare); theoretically, same adverse effects as ACE inhibitors on renal function (except proteinuria and neutropenia), serum potassium, and blood pressure
Candesartan	
Irbesartan	
Losartan	
Olmesartan	
Telmisartan	
Valsartan	

* All ACE inhibitors and angiotensin II receptor blockers are contraindicated in pregnancy (can cause injury or death to the developing fetus).

ACE inhibitors references

1. Palla M, Ando T, Androulakis E, Telila T, Briasoulis A. Renin-Angiotensin System Inhibitors vs Other Antihypertensives in Hypertensive Blacks: A Meta-Analysis. *J Clin Hypertens (Greenwich).* 2017;19(4):344-350. doi:10.1111/jch.12867
2. Israili ZH, Hall WD. Cough and angioneurotic edema associated with angiotensin-converting enzyme inhibitor therapy. A review of the literature and pathophysiology. *Ann Intern Med.* 1992;117(3):234-242. doi:10.7326/0003-4819-117-3-234
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4. Townsend RR, Holland OB. Combination of converting enzyme inhibitor with diuretic for the treatment of hypertension. *Arch Intern Med.* 1990;150(6):1175-1183.
5. Lacourcière Y, Poirier L, Lefebvre J, Ross SA, Leenen FH. Increasing the doses of both diuretics and angiotensin receptor blockers is beneficial in subjects with uncontrolled systolic hypertension. *Can J Cardiol.* 2010;26(8):313-319. doi:10.1016/s0828-282x(10)70442-6

Angiotensin II Receptor Blockers for Hypertension

Angiotensin II receptor blockers (see table Oral ACE Inhibitors and Angiotensin II Receptor Blockers for Hypertension) block angiotensin II receptors and therefore interfere with the renin-angiotensin system. Angiotensin II receptor blockers and ACE inhibitors are equally effective as antihypertensives. Angiotensin II receptor blockers may provide added benefits via tissue ACE blockade. The 2 classes have the same beneficial effects in patients with left ventricular failure or with nephropathy due to type 1 diabetes. An angiotensin II receptor blocker should not be used together with an ACE inhibitor, but when used with a beta-blocker may reduce the hospitalization rate for patients with heart failure. Angiotensin II receptor blockers may be safely used in anyone with an estimated GFR > 30 mL/minute to reduce cardiovascular risk and kidney disease progression.

Incidence of adverse events is low; angioedema occurs but much less frequently than with ACE inhibitors. Precautions for use of angiotensin II receptor blockers in patients with renovascular hypertension, hypovolemia, and severe heart failure are the same as those for ACE inhibitors (see table Oral ACE Inhibitors and Angiotensin II Receptor Blockers for Hypertension). Angiotensin II receptor blockers are contraindicated during pregnancy.

Calcium Channel Blockers for Hypertension

Dihydropyridines (see table Oral Calcium Channel Blockers for Hypertension) are potent peripheral vasodilators and reduce blood pressure by decreasing total peripheral vascular resistance (TPR); they sometimes cause reflexive tachycardia.

The **nondihydropyridines** verapamil and diltiazem slow the heart rate, decrease atrioventricular conduction, and decrease myocardial contractility. These medications should not be prescribed for patients with second-degree or third-degree atrioventricular block or with left ventricular failure.

TABLE

Oral Calcium Channel Blockers for Hypertension

Medication	Selected Adverse Effects	Comments
Dihydropyridines		

Medication	Selected Adverse Effects	Comments
Amlodipine	Dizziness, flushing, headache, weakness, nausea, heartburn, peripheral edema, tachycardia Possible gingival hyperplasia with chronic use	Contraindicated in heart failure, possibly except for amlodipine Use of short-acting nifedipine possibly associated with higher rate of myocardial infarction
Felodipine		
Isradipine		
Nicardipine		
Nicardipine, sustained-release		
Nifedipine, extended-release		
Nisoldipine		
Nondihydropyridines		
Diltiazem, sustained-release	Headache, dizziness, asthenia, flushing, peripheral edema, bradycardia; possibly liver dysfunction Possible gingival hyperplasia with chronic use Verapamil: Constipation	Contraindicated in heart failure with reduced ejection fraction due to negative inotropic effects, in sinus node dysfunction, or in greater than first-degree atrioventricular block
Diltiazem, extended-release		
Verapamil		
Verapamil, sustained-release		

Use of short-acting nifedipine should be avoided because of an increased risk of acute myocardial infarction (1).

A calcium channel blocker is preferred to a beta-blocker in patients with angina pectoris who also have a bronchospastic disorder, with coronary spasms, or with Raynaud phenomenon.

Calcium channel blocker reference

1. Furberg CD, Psaty BM, Meyer JV.Nifedipine. Dose-related increase in mortality in patients with coronary heart disease. *Circulation.* 1995;92(5):1326-1331. doi:10.1161/01.cir.92.5.1326

Direct Renin Inhibitor for Hypertension

Aliskiren, a direct renin inhibitor, is used in the management of hypertension.

As with ACE inhibitors and angiotensin II receptor blockers, aliskiren causes elevation of serum potassium and creatinine. Aliskiren should not be combined with ACE inhibitors or angiotensin II receptor blockers in patients with diabetes or renal disease (estimated GFR < 60 mL/minute). It is also contraindicated during pregnancy.

Beta-Blockers for Hypertension

Beta-blockers are not first-line agents for treatment of hypertension. However, they may be useful in patients with hypertension who have other disorders that may benefit from a beta-blocker, such as angina, previous myocardial infarction, or heart failure. Otherwise, beta-blockers are less protective against stroke and overall mortality than some other antihypertensives (1, 2).

Beta-blockers (see table Oral Beta-Blockers for Hypertension) slow the heart rate and reduce myocardial contractility, thus reducing blood pressure. All beta-blockers are similar in antihypertensive efficacy. Cardioselective beta blockers (eg, acebutolol, atenolol, betaxolol, bisoprolol, metoprolol) are often preferred over nonselective agents because of potentially less bronchoconstriction and peripheral vasodilation, which is particularly relevant for patients with diabetes (increasing risk of hypoglycemia), chronic peripheral artery disease (impairing function), or chronic obstructive pulmonary disease (COPD, by potentiating bronchospasm). However, cardioselectivity is only relative and decreases as dose increases. Even cardioselective beta-blockers should be used with caution in patients with COPD with a prominent bronchospastic component.

TABLE

Oral Beta-Blockers for Hypertension

Medication	Selected Adverse Effects	Comments
Acebutolol*, †	Bronchospasm, fatigue, insomnia, sexual dysfunction, exacerbation of heart failure, masking of symptoms of hypoglycemia, triglyceridemia, increased total cholesterol and decreased high-density lipoprotein cholesterol (except for pindolol,	Contraindicated in patients with greater than 1st-degree atrioventricular block, or sick sinus syndrome Should be avoided in patients with asthma Should be used cautiously in patients with COPD with severe disease, heart
Atenolol*		
Betaxolol*		
Bisoprolol*		
Carvedilol‡		
Carvedilol (controlled-release)‡		
Labetalol‡, §		

Medication	Selected Adverse Effects	Comments
Metoprolol*	acebutolol, carteolol, and labetalol)	failure, or who are taking insulin Should not be stopped abruptly in patients with coronary artery disease Bisoprolol, carvedilol, and metoprolol can be used to treat heart failure
Metoprolol (extended-release)		
Nadolol		
Nebivolol		
Pindolol†		
Propranolol		
Propranolol, long-acting		
Timolol		

* Cardioselective.

† With intrinsic sympathomimetic activity.

‡ Alpha-beta-blockers. Labetalol can also be given IV for hypertensive emergencies.

§ Can also be given for hypertensive emergencies.

Beta-blockers with intrinsic sympathomimetic activity (eg, acebutolol, pindolol) do not adversely affect serum lipids; they are less likely to cause severe bradycardia.

Beta-blockers have central nervous system (CNS) adverse effects (sleep disturbances, fatigue, lethargy) and exacerbate depression. Nadolol affects the CNS the least and may be best when CNS effects must be avoided. Beta-blockers are contraindicated in patients with second- or third-degree atrioventricular block or sinus node dysfunction. Beta-blockers should generally be avoided in patients with asthma because, in addition to bronchospasm, they can also cause resistance to the effects of inhaled or oral beta receptor agonists (3).

Beta-blockers references

1. Thomopoulos C, Bazoukis G, Tsioufis C, Mancia G. Beta-blockers in hypertension: overview and meta-analysis of randomized outcome trials. *J Hypertens.* 2020;38(9):1669-1681. doi:10.1097/HJH.0000000000002523
2. Wiysonge CS, Bradley HA, Volmink J, Mayosi BM, Opie LH. Beta-blockers for hypertension. *Cochrane Database Syst Rev.* 2017;1(1):CD002003. doi:10.1002/14651858.CD002003.pub5

3. Morales DR, Jackson C, Lipworth BJ, Donnan PT, Guthrie B. Adverse respiratory effect of acute β-blocker exposure in asthma: a systematic review and meta-analysis of randomized controlled trials. *Chest.* 2014;145(4):779-786. doi:10.1378/chest.13-1235

Alpha Adrenergic Modifiers for Hypertension

Adrenergic modifiers include central alpha-2-agonists, postsynaptic alpha-1-blockers, and peripheral-acting non-selective adrenergic blockers (see table Adrenergic Modifiers for Hypertension).

TABLE

Alpha Adrenergic Modifiers for Hypertension

<table>
<tr><th>Medication*</th><th>Selected Adverse Effects</th><th>Comments</th></tr>
<tr><td colspan="3">Alpha-2-agonists (central acting)</td></tr>
<tr><td>Clonidine</td><td rowspan="4">Drowsiness, sedation, dry mouth, fatigue, sexual dysfunction, rebound hypertension with abrupt discontinuance (particularly if doses are high or concomitant beta-blockers are continued), localized skin reaction to clonidine patch; possibly liver damage, Coombs-positive hemolytic anemia with methyldopa</td><td rowspan="4">Should be used cautiously in older patients because of orthostatic hypotension
Interferes with measurements of urinary catecholamine levels by fluorometric methods
Should not be combined with an alpha-1-blocker because of a risk of significant orthostatic hypotension.</td></tr>
<tr><td>Clonidine TTS (patch)</td></tr>
<tr><td>Guanfacine</td></tr>
<tr><td>Methyldopa</td></tr>
<tr><td colspan="3">Alpha-1 blockers</td></tr>
<tr><td>Doxazosin</td><td rowspan="3">First-dose syncope, orthostatic hypotension, weakness, palpitations, headache</td><td rowspan="3">Should be used cautiously in older patients because of orthostatic hypotension
Relieves symptoms of benign prostatic hyperplasia</td></tr>
<tr><td>Prazosin</td></tr>
<tr><td>Terazosin</td></tr>
</table>

* Peripheral-acting adrenergic blockers (eg, guanadrel, guanethidine, reserpine) are not available in the United States.

TTS = transdermal therapeutic system.

Alpha-2-agonists (eg, methyldopa, clonidine, guanfacine) stimulate alpha-2-adrenergic receptors in the brain stem and reduce sympathetic nervous activity, lowering blood pressure (BP). Because they have a central action, they are more likely than other antihypertensives to cause drowsiness, lethargy, and depression; they are not widely used. Clonidine can be applied transdermally once a week as a patch; thus, it may be useful for patients who have difficulty adhering to treatment (eg, those with dementia). Rebound hypertension can occur with abrupt discontinuation of oral clonidine.

Postsynaptic alpha-1-blockers (eg, prazosin, terazosin, doxazosin) are not used for primary treatment of hypertension because of their adverse effects and no reduction in mortality. Doxazosin used alone or with antihypertensives other than diuretics increases the risk of heart failure. Other adverse effects include first-dose syncope, orthostatic hypotension, weakness, palpitations, and headache. However, postsynaptic alpha-1-blockers may be used in patients who have prostatic hypertrophy and need a fourth antihypertensive or in people with high sympathetic tone (ie, with high heart rate and spiking blood pressures) already on the maximum dose of a beta-blocker.

Direct Vasodilators for Hypertension

Direct vasodilators, including minoxidil and hydralazine (see table Direct Vasodilators for Hypertension), work directly on blood vessels, independently of the autonomic nervous system. Minoxidil is more potent than hydralazine but has more adverse effects, including sodium and water retention and hypertrichosis, which may be poorly tolerated. Minoxidil should be reserved for severe, refractory hypertension.

Hydralazine may be used during pregnancy (eg, for preeclampsia) and as an adjunct antihypertensive. Hydralazine (particularly at doses > 200 mg/day) has been associated with drug-induced lupus, which resolves when the medication is stopped (1). It is also associated with drug-induced antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis (2).

TABLE

Direct Vasodilators for Hypertension

Medication	Selected Adverse Effects*	Comments
Hydralazine	Positive antinuclear antibody test, drug-induced lupus (rare at recommended doses)	Augments vasodilating effects of other vasodilating medication
Minoxidil	Sodium and water retention, hypertrichosis; possibly new or worsening pleural and pericardial effusions	Reserved for severe, refractory hypertension

* Both medications may cause headache, tachycardia, and fluid retention and may precipitate angina in patients with coronary artery disease.

Direct vasodilators references

1. Handler J. Hydralazine-induced lupus erythematosis. *J Clin Hypertens (Greenwich).*2012;14(2):133-136. doi:10.1111/j.1751-7176.2011.00573.x
2. Santoriello D, Bomback AS, Kudose S, et al. Anti-neutrophil cytoplasmic antibody associated glomerulonephritis complicating treatment with hydralazine. *Kidney Int.* 2021;100(2):440-446. doi:10.1016/j.kint.2021.03.029

Drug Information for the Topic